

Towards a structure theory for rank two Kleinian groups

Extended abstract for Oberwolfach report

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A Kleinian group is a discrete subgroup of $\mathrm{PSL}(2, \mathbb{C}) \simeq \mathrm{Isom}^+(\mathbb{H}^3) \simeq \mathrm{Möb}^+(\mathbb{S}^3)$. If Γ is a Kleinian group, then Γ acts discontinuously on $\mathbb{H}^3$ and $\mathbb{H}^3/\Gamma$ is a hyperbolic orbifold. Further, Γ acts discontinuously on an open subset $\Omega(\Gamma) \subset \mathbb{S}^2 = \partial\mathbb{H}^3$, and $\Omega(\Gamma)/\Gamma$ is a (possibly empty, possibly disconnected, possibly punctured) Riemann surface.

Suppose that $\eta : \mathbb{C}^n \rightarrow \mathrm{PSL}(2, \mathbb{C})^m$ is a holomorphic parameterisation of m -sets of matrices with n complex parameters, such that $\langle \eta(\mathbf{0}) \rangle$ is Kleinian. (We have the holomorphism condition since when $n = 1$ this allows for the use of the rich theory of holomorphic motions [2, Chapter 12]; higher-dimensional analogues of this are not well-understood.) The set

$$\{\mathbf{z} \in \mathbb{C}^n : \langle \eta(\mathbf{z}) \rangle \text{ is discrete and } \mathbb{H}^3/\langle \eta(\mathbf{z}) \rangle \simeq_{\text{homeo.}} \mathbb{H}^3/\langle \eta(\mathbf{0}) \rangle\}$$

is very wild in general; for instance, not locally connected in the subspace topology from $\mathbb{C}^n$ (see e.g. [12]).

We now restrict to rank two groups. First suppose both generators are elliptic or parabolic. If the orders are $a, b \in \mathbb{N} \cup \{\infty\}$ then up to conjugacy there is a one-parameter family of such generating sets:

$$\Gamma_z = \left\langle \begin{bmatrix} e^{i\pi/a} & 1 \\ 0 & e^{-i\pi/a} \end{bmatrix}, \begin{bmatrix} e^{i\pi/b} & 0 \\ z & e^{-i\pi/b} \end{bmatrix} \right\rangle.$$

The set of all z such that Γ_z is discrete has been completely classified. The parabolic-parabolic discrete groups are classified by Keen and Series [9], Maskit and Swarup [11], and Ohshika and Miyachi [13] (free groups) and by Akiyoshi, Ohshika, Parker, Sakuma, Yoshida [1] (non-free groups). The groups with possibly elliptic generators have been classified by Elzenaar, Martin, and Schillewaert [4, 8] and Chesebro, Martin and Schillewaert [3]. Essentially, there is one large open set $\mathcal{R}$ of flexible discrete groups which is a quotient of a Teichmüller space of four-times marked spheres; the closure of this set parameterises all Γ_z which are discrete and isomorphic to $\mathbb{Z}/a\mathbb{Z} * \mathbb{Z}/b\mathbb{Z}$. In addition

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there are isolated discrete groups, either lying on the real axis or accumulating onto $\partial\mathcal{R}$. The groups on the real line are classified by standard results on Fuchsian groups [14, 10]. All the others are virtually (up to index 2) holonomy groups of orbifold structures on 2-bridge link complements with possibly singular unknotting tunnels. In particular, if a group Γ_z is discrete, torsion-free, non-elementary, and non-Fuchsian, then it is either a 2-bridge link group or lies in $\mathcal{R}$. One application of this work is to computing the radius of convergence of q -rational numbers [7].

One can ask how much of this picture generalises to the general rank two case. It is easy to make some naïve conjectures, of which here are two:

Conjecture A. *There exists a Keen–Series pleating ray theory for genus two Schottky space.* The case of rational pleating rays is discussed in my recent preprint [5]. In an upcoming new version of this preprint, a description of irrational pleating rays will also be given (arising from recent conversations with Bruno Dular) which will complete the picture of the coordinate system, modulo understanding the asymptotic behaviour of the pleating rays which relies (in the one-dimensional case) on explicit estimates which are not available in general.

Conjecture B. *If α, β, γ are disjoint simple closed curves on the boundary of a genus two handlebody and $\rho : F_2 \rightarrow \mathrm{PSL}(2, \mathbb{C})$ is a discrete, non-elementary, non-Fuchsian representation of the fundamental group of the genus two handlebody such that*

$$\mathrm{tr}^2 \rho(\alpha) = \mathrm{tr}^2 \rho(\beta) = \mathrm{tr}^2 \rho(\gamma) = 4, \quad (*)$$

then either:

1. *If all three of $\rho(\alpha), \rho(\beta), \rho(\gamma)$ are parabolic, then $\Gamma = \rho(F_2)$ is a maximal cusp group on the boundary of genus two Schottky space and $\Omega(\Gamma) \neq \emptyset$.*
2. *If exactly two are parabolic, then Γ is a tunnel number 1 link group in some ambient closed 3-manifold.*
3. *If 0 or 1 are parabolic, then Γ is a Dehn filling of a group appearing in 2.*

In recent work [6] I show that 1. of Conjecture B is false. There exists a set of three curves α, β, γ and a discrete representation ρ such that $\rho(\alpha), \rho(\beta), \rho(\gamma)$ are all parabolic but $\rho(F_2)$ acts ergodically on the 2-sphere, i.e. $\Omega(\Gamma) = \emptyset$. Thus, unlike the case of elliptic or parabolic generators, the assumption $\Omega \neq \emptyset$ is necessary to deduce that a group realising three disjoint curves on the boundary of the genus 2 handlebody by parabolics is a maximal cusp group.

In upcoming work, I will give a structure theory for the groups which appear in 2. and 3. of Conjecture B, i.e. tunnel number 1 link groups and their Dehn fillings. This is a first step to understanding the general solution set of (*).

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